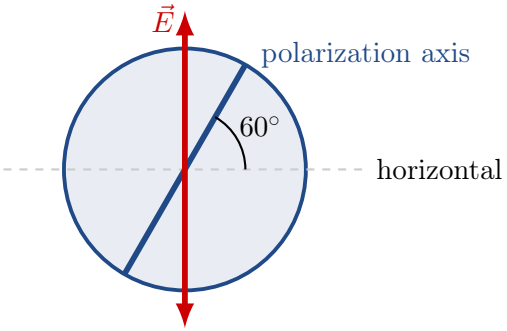
1. The electric field of a light wave travelling through a block of clear material is

$$E(x,t) = 50 \sin\left(1.5 \times 10^7\, x - 3 \times 10^{15}\, t\right)$$

A. How fast is this wave moving? What is the index of refraction n of the material?

B. The block is 6.0 cm thick. How long does the light take to cross it?

2. A **vertically polarized** laser beam of intensity $I_0 = 800$ W/m^2 hits a polarizing filter head-on. The filter's axis is tilted 60° up from **horizontal**, as shown.

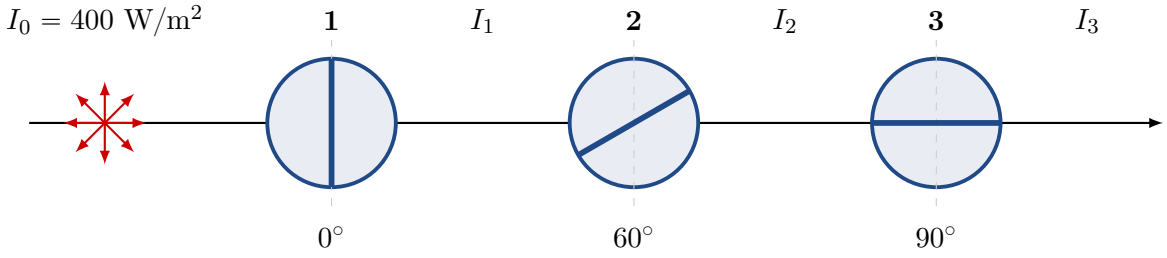


A. What angle θ belongs in Malus's law here? Mark it on the figure.

B. What is the intensity of the beam after the filter?

C. On the figure, draw the direction of $\vec{E}$ **after** the filter.

3. Unpolarized light of intensity $I_0 = 400$ W/m^2 passes through three polarizers in a row. Their axes are tilted 0°, 60° and 90° from vertical.



A. Find the intensity after each polarizer.

$I_1 =$ _____ $I_2 =$ _____ $I_3 =$ _____

B. On the figure, draw the direction of $\vec{E}$ in each of the three gaps after a polarizer.

C. Polarizer **2** is pulled out. Now how much light gets through? Why did taking a filter *away* make it darker?